The Stochastic Smoluchowski Kramers Approximation with a Mean Field Nonlinearity

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Motivation: The Stochastic Quantization Program

Goal:

Sample from a euclidean quantum field theory described via a target measure

$$d\mu[\phi] = \frac{1}{Z} e^{-S_E[\phi]} \, d\mathcal{D}\phi.$$

Challenge:

Direct sampling from μ is highly intractable.

Strategy: Stochastic Quantization

Construct stochastic Langevin dynamics ϕ_τ such that:

1. **Invariance:** μ is the unique equilirbium state.
2. **Convergence:** For an easy-to-sample initial distribution $\phi_0 \sim \rho$,

$$P(\phi_\tau) \xrightarrow{\tau \to \infty} \mu[\phi].$$

$\Rightarrow$ *Sample from μ by computing ergodic averages of ϕ_τ for large Burn-in times.*

Background: The Hyperbolic $O(N)$ Φ^4_2-Model

The target Euclidean QFT on $\mathbb{T}^2$ is reached by sampling the first marginal of:

$$d\rho_N[\phi_N, \pi_N] = \frac{1}{Z_N} \exp\left[-S_E[\phi_N] - \frac{1}{2}\int_{\mathbb{T}^2} \sum_{j=1}^{N} |\pi_{N,j}|^2 dx\right] \mathcal{D}\phi_N \, \mathcal{D}\pi_N$$

where $\pi_N = \partial_\tau \phi_N$ is the auxiliary fictitious momentum, and the Euclidean action is:

$$S_E[\phi_N] = \int_{\mathbb{T}^2} \left(\frac{1}{2} \sum_{j=1}^{N} (|\nabla \phi_{N,j}|^2 + |\phi_{N,j}|^2) + \frac{1}{4N} : \Big(\sum_{j=1}^{N} \phi_{N,j}^2 \Big)^2 : \right) dx.$$

Hyperbolic Stochastic Quantization

[Tolomeo, 2023], [Liu, 2025]: ρ_N is the unique equilibrium state of ($1 \leq j \leq N$):

$$\partial_\tau^2 \phi_{N,j} + \partial_\tau \phi_{N,j} + (1 - \Delta)\phi_{N,j} + \frac{1}{N} \sum_{k=1}^{N} :(\phi_{N,k})^2 \phi_{N,j}: = \sqrt{2}\xi(\tau, x),$$

where $\xi(\tau, x)$ is space-(fictitious)-time white noise.

The Mean-Field Limit

The Stochastic Damped Mean Field Wave Equation (SDMFW)

[Liu, 2025]: As $N \to \infty$, the interactions vanish and the target measures ρ_N converges to the non-interacting Gaussian measure $\mu_0 \otimes \mu_1$. The corresponding underdamped Langevin dynamics converges to:

$$\partial_\tau^2 \phi + \partial_\tau \phi + (1 - \Delta)\phi + \langle :\phi^2: \rangle \phi = \sqrt{2}\xi(\tau, x).$$

Theorem 1 (Contribution 1: Unique Equilibrium of the SDMFW)

Let $\sigma \in (0, \frac{1}{2})$. The Gaussian measure $\mu_0 \otimes \mu_1$ is the unique equilibrium state of the SDMFW in the class of measures with a bounded normal-ordered second moment:

$$\langle \int_{\mathbb{T}^2} |\nabla^{-\sigma} :\phi^2: |^2 \, dx \rangle < \infty.$$

Smoluchowski-Kramers Approximation (SKA)

Context [Zine, 2024; Liu, 2026]

The $\mathrm{HLSM_N}$ converges to the Parabolic $O(N)$ Linear Sigma Model ($\mathrm{PLSM_N}$) (the transition from underdamped to overdamped Langevin dynamics). We establish the analogous result for their mean-field limits.

Theorem 2 (Contribution 2: From Wave to Heat: SDMFW $\to$ SMFH)

As the fictitious inertia $\varepsilon \to 0$, the damped wave equation (SDMFW) converges globally to the heat equation (SMFH). Concurrently, the fictitious momentum decouples, and converges to 0*:*

$$\varepsilon^2 \partial_\tau^2 \phi_\varepsilon + \partial_\tau \phi_\varepsilon + (1-\Delta)\phi_\varepsilon + \langle :\phi_\varepsilon^2: \rangle \phi_\varepsilon = \sqrt{2}\xi(\tau, x)$$

$$\Downarrow \quad \varepsilon \to 0$$

$$\partial_\tau \phi_0 + (1-\Delta)\phi_0 + \langle :\phi_0^2: \rangle \phi_0 = \sqrt{2}\xi(\tau, x)$$

$$d\mu_0 \otimes d\mu_\varepsilon$$

$$\varepsilon \to 0 \quad \Downarrow$$

$$d\mu_0$$

Summary:

The following diagram summarizes the relationships between the $O(N)$ models, their mean-field limits, and the respective SKA.

$$
\begin{tikzcd}
\boxed{\mathrm{HLSM}_N} \arrow[r, "N\to\infty", "\text{[Liu, 2025]}"'] \arrow[d, "\varepsilon\to0"', "\text{[Liu, 2026]}"] & \boxed{\mathrm{SDMFW}} \arrow[d, "\varepsilon\to0"] \\
\boxed{\mathrm{PLSM}_N} \arrow[r, "\text{[Shen, 2022]}", "N\to\infty"'] & \boxed{\mathrm{SMFH}}
\end{tikzcd}
$$

Figure: Diagram of limits between the Hyperbolic and Parabolic models.

References I

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